RUIQI DING (22 YO)

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WORKING EXPERIENCE

Reinforcement Learning Quantum Control

Yvonne Y. Gao (MIT TR35) group, CQT

Jul 2020 – Ongoing quantumcrew.org

- We design novel quantum circuits, develop robust hardware for quantum computers, probe quantum effects at 10mK, and build test-beds for quantum algorithms.
- Currently, I am working on quantum optimal control theory using the state of the art **deep reinforcement learning** algorithms.
- Tech stack: PyTorch, Autograd (Machine learning)
- Qcrew/Reinforcement-Learning-Quantum-Control

RESEARCH PROJECTS

Reinforcement Learning Nanobot

Reiner Kree group, University of Göttingen

Apr 2019 – Aug 2019

- We study the Statistical Physics of disordered and complex systems such as: biomembranes, cell cultures, neural networks and biological neurons.
- Specifically, we successfully trained nanobots to navigate through non-observable complex fluids such as human tissues using **deep reinforcement learning**. The fluid environment was characterised by Markovian transition matrices so that the nanobot undergoes random walks at each time step.
- Tech stack: PyTorch (Machine learning)
- tesla-cat/Reinforcement-Learning-Nanobot

FPGA Trapped Ion Control

Dzmitry Matsukevich (h-index: 30) group, CQT

Aug 2019 – Dec 2019 dzmitrylab.quantumlah.org

- We use trapped Ytterbium ions to study quantum information processing and the implementation of heat machines in the quantum regime, as well as precision measurements with atomic and molecular ions.
- Specifically, I worked on FPGA programming using the state of the art Cyclone V SoC FPGAs. The FPGA is used to generate sequencing signals to control the laser pulse that drives trapped ions, it is also used for photon counting. In addition, I developed a full-stack framework that handles the communication between FPGA hardware and Python programs.
- Tech stack: FPGA(Verilog), C, Node.js, Python
- tesla-cat/FPGA-Trapped-Ion-Control

EDUCATION

M.Sc. in Physics

University of Göttingen, Germany

Apr 2019 – Aug 2019

- GPA: mine is 1.4 (Very good in German system), best is 1.0, **pass** is 4.0
- Transcript: 3-uni-goettingen-grades.pdf
- This was supposed to be a bachelor exchange program, but I liked challenges, so I took master courses in Biophysics.

B.Sc. in Physics with Honours (Highest Distinction)

National University of Singapore

Aug 2016 – May 2020

- GPA: mine is 4.52 (Highest Distinction), best is 5.0, worst is 0
- Degree Scroll: 2-NUS-Degree.pdf
- Transcript: 2-NUS-Transcript.pdf
- Thesis (grade: 5.0): Density Potential Functional Theory in position and momentum space and its implementation using the Machine Learning library PyTorch.

CODING SKILLS

Python | PyTorch (Machine learning) | NumPy, SciPy etc. (Data Analysis) | JavaScript/TypeScript | React.js (Web dev) | React Native (Mobile Dev) | C++ | Arduino (Embedded systems) | ESP32 (IoT) | Verilog (FPGA) | MATLAB | Mathematica

OTHER SKILLS

Physics and Math (analytical) | Altium (electronics design) | SolidWorks (mechanical design) | Linux, PowerShell

Density Potential Functional Theory (Bachelor Thesis)

Berge Englert (h-index: 51) group, CQT

Aug 2019 – May 2020

- We study low-dimensional ultracold dilute gases of fermions and bosons, the band structure, transport properties, and finite-temperature effects of many-body systems in periodic lattices.
- Specifically, I worked on the Density Potential Functional Theory (DPFT) established by Julian Schwinger (Nobel Prize in Physics, 1965) and his postdoc Berge Englert. Its application on spin polarized Fermi gas with magnetic dipole-dipole interaction was studied under the Thomas Fermi approximation in both position and momentum space. The code used the **Machine Learning** library PyTorch to achieve multi-GPU acceleration.
- Tech stack: PyTorch (GPU acceleration, parallelism)
- tesla-cat/PyDPFT

PROJECTS

Lab Kitten

- Lab Kitten is a toolkit for labs / experiments using real-time communication technology, it enables researchers to monitor lab data and control lab instruments conveniently no matter where they are, for example at home or on a bus.
- Specifically, it uses WebRTC technology, which is the real-time technology of highest performance and has been widely used in live-streaming, video chat applications etc.
- Tech stack: WebRTC , Python , Node.js , socket , React.js , Plotly.js
- lab-kitten/lab-kitten

FPGA-IoT Musical Tesla Coil

- The goal is to build a Musical Tesla Coil using two advanced technologies: IoT and FPGA.
- A front-end UI was written in Javascript, it parses MIDI music files to extract frequencies and volumes, then sends them to the ESP32 IoT module that runs a websocket server, which in turn sends SPI and PWM signals to control the FPGA. The FPGA handles controlled phase shifting to drive the IGBT Full bridge inverter, which in turn drives the Tesla Coil's LC resonator to produce high-voltage plasma music. All PCBs were designed and soldered by myself.
- Tech stack: FPGA(Verilog) , ESP32 IoT , C++ , WebSocket , JavaScript
- tesla-cat/FPGA-ESP32-Musical-Tesla-Coil
- 1-musical-tesla-coil.mp4

AWARDS

China-Singapore intergovernmental scholarship (SM2)
Only 90 quota each year in the entire China, only available to best students in best high schools.

National University of Singapore Science and Technology Undergraduate Scholarship
Continuation of the above, awarded to those who pass one year of observation period.

China's nationwide high school physics competition third prize

HUAWEI High School Science Fair best performance award

OTHERS

- **Citizenship**: People's Republic of China
- **Languages**: Chinese, English